

# A Stopping Target Optimized at the *Deployed* Foil Pitch

9D shape search + a pitch-profile axis, inside the deployed envelope

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Mu2e — autoresearch / closed-loop Bayesian optimization

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# Result in one slide

Hold the stopping target inside the **deployed envelope** (extent  $48 \times 22.222$  mm) and let Bayesian optimization choose everything else. At the *deployed damage budget* the best design gives +23%  $S/\sqrt{B}$  (4.00 vs 3.26); with no damage cost at all, +19% **at** -13% **damage**. Unconstrained, the significance record is **4.41** (+35%) at +60% damage — and still rising.

- **297 evaluations** (13 transplanted seeds + 284 new) across ten campaigns — shape, widened box, pitch profile, and two  $\max S/\sqrt{B}$  exploit rounds — **zero failures**. Search **closed for deployment**; the exploit rounds map the unconstrained ceiling.
- **Deployment answer** bp04R00\_09: **4.00** at  $6.37 \times 10^{-7}$  — the most significance available *within* the deployed target's damage (three campaigns confirmed nothing beats it there).
- **No-cost option** bp04R00\_02: +19%  $S/\sqrt{B}$  **at** -13% **damage** — strictly better than the deployed target on *both* axes. **42 of 297** designs do this.
- **Record** foilspfbpz05R01\_00: **4.41** at  $1.10 \times 10^{-6}$  — +35%  $S/\sqrt{B}$  for +60% damage. The exploit record trajectory  $4.33 \rightarrow 4.37 \rightarrow \mathbf{4.41}$  ( $\sim 5\sigma$  steps at the replicate-measured  $\sigma_{\text{sob}} = 0.006$ ) had *not* flattened at stop.
- Damage frontier spans **23** $\times$ , floor  $5.6 \times 10^{-8}$  MeV/POT, **41 measured Pareto points**.

All numbers Run1Bap-scale on the corrected stack (IPA-position fix, strict zero-overlap preflight).

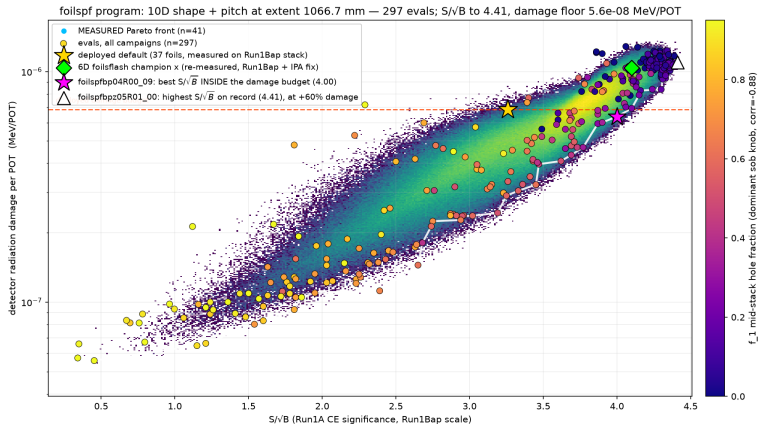
# What we optimize — and what we deliberately do not

**9-D shape at fixed pitch** (later +1): 49 foils generated from  $K=3$  quadratic control points for outer radius, half-thickness and hole fraction. **Extent is pinned** at  $48 \times 22.222 = \mathbf{1066.7}$  mm — the deployed foil spacing. Late campaigns widened the `rOut` box to 30–150 and added `zmid` (a linearly-varying foil pitch; `zmid = 0 = uniform`).

| knob                      | range                          |               | objective                         | direction       |
|---------------------------|--------------------------------|---------------|-----------------------------------|-----------------|
| <code>rOut_{0,1,2}</code> | 50–120 ( $\rightarrow$ 30–150) | mm            | $S/\sqrt{B}$ (Run1A CE)           | <b>maximize</b> |
| <code>hT_{0,1,2}</code>   | 0.01–0.15                      | mm            | detector radiation damage per POT | <b>minimize</b> |
| <code>f_{0,1,2}</code>    | 0–0.95                         | hole fraction |                                   |                 |
| <code>zmid</code> (late)  | –150–+150                      | mm            |                                   |                 |
| extent                    | <b>1066.7</b>                  | <b>fixed</b>  |                                   |                 |

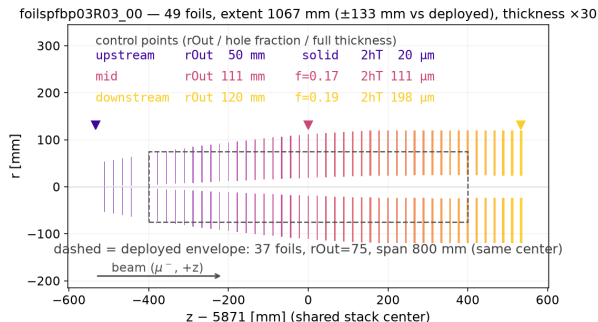
**Why fix the pitch.** A designed extent scan (10 evaluations, two shapes, 800–2000 mm) showed  $S/\sqrt{B}$  flattening by 1400–1700 mm: the last 300 mm buys +0.9% significance for +8% damage. Length is a settled question — so spend the search on shape, inside an envelope the hardware already supports.

# The landscape



All **297** evaluations; GP density cloud over the full 10D box (background). The white line is the **measured** Pareto front (41 points) — not a GP prediction. Dashed line = the deployed damage budget: magenta star = best design *inside* it (4.00); **white triangle** = the unconstrained record (4.41, +60% damage); gold star = deployed default; green diamond = 6D foilsflash champion re-measured. Colour = mid-stack hole fraction, the dominant knob (corr **-0.88**).

# Anatomy of the top- $S/\sqrt{B}$ designs



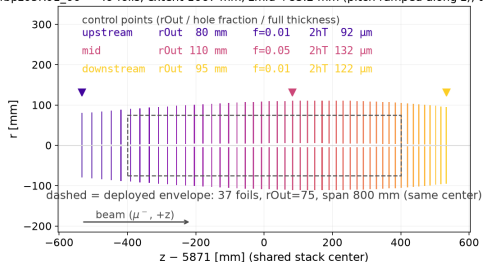
| design                          | $S/\sqrt{B}$ | damage      |
|---------------------------------|--------------|-------------|
| bpz05R01_00 (record)            | <b>4.41</b>  | 1.10        |
| bpz04R11_00                     | 4.37         | 1.04        |
| bp03R03_00 (sketched)           | 4.30         | <b>0.84</b> |
| bp02R08_00                      | 4.05         | <b>0.73</b> |
| 6D champion $\times$ (re-meas.) | 4.10         | 1.04        |
| deployed default                | 3.26         | 0.69        |

damage in  $10^{-6}$  MeV/POT, all on the Run1Bap stack

- **Sketch: bp03R03\_00. Grows outward, stays nearly solid.** rOut 50/111/120 mm with holes almost shut ( $f = 0/0.17/0.19$ ); thickness ramps  $20 \rightarrow 198 \mu\text{m}$  — thin where the beam is intact, thick where it is spent.
- **The record family (next slide): 4.37** (bpz04R11\_00) is a wide funnel-bore stack at *uniform* pitch, entirely inside the original 9D box; **4.41** (bpz05R01\_00) is a *near-solid* barrel at ramped pitch. Both found only by the dedicated exploit rounds — the hypervolume picker never

# The best- $S/\sqrt{B}$ target design

foilsfpbpz05R01\_00 — 49 foils, extent 1067 mm, zmid +83.2 mm (pitch ramped along z), thickness  $\times 30$



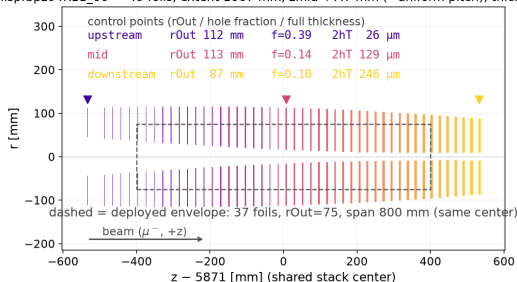
| design               | $S/\sqrt{B}$ | damage      |
|----------------------|--------------|-------------|
| bpz05R01_00 (record) | <b>4.41</b>  | 1.10        |
| deployed default     | 3.26         | <b>0.69</b> |
| <i>difference</i>    | +35%         | +60%        |

damage in  $10^{-6}$  MeV/POT, Run1Bap stack

- **Nearly solid.** Bore radius  $\leq 5$  mm everywhere ( $f = 0.01/0.05/0.01$ ) — no funnel, no hole: all stopping mass kept.
- **A gentle barrel, heavy throughout:** rOut 80  $\rightarrow$  110  $\rightarrow$  95 mm; thickness 92/132/122  $\mu\text{m}$  — not the thin-to-thick ramp of the other top designs.
- **Pitch ramped:** zmid = +83 mm — foils sparse upstream, packed downstream (visible in the sketch).
- **A different family** from the 4.37 funnel (next slide): the ridge top is degenerate — two dissimilar shapes within 1% in  $S/\sqrt{B}$ .

# The 4.37 family — a funnel bore at uniform pitch

foilsfpbpz04R11\_00 — 49 foils, extent 1067 mm, zmid +7.7 mm ( $\approx$ uniform pitch), thickness  $\times 30$



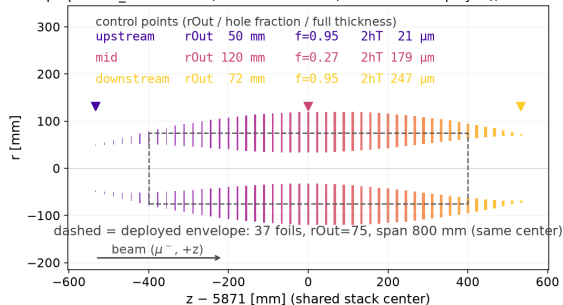
| design                     | $S/\sqrt{B}$ | damage      |
|----------------------------|--------------|-------------|
| bpz04R11_00 (prev. record) | <b>4.37</b>  | 1.04        |
| deployed default           | 3.26         | <b>0.69</b> |
| <i>difference</i>          | +34%         | +52%        |

damage in  $10^{-6}$  MeV/POT, Run1Bap stack

- **A wide barrel with a funnel bore.** rOut starts wide and stays wide (112  $\rightarrow$  113  $\rightarrow$  87 mm) while the hole **closes along the beam**: bore radius 44  $\rightarrow$  16  $\rightarrow$  9 mm — open where the beam enters, nearly solid where it is spent.
- Thickness ramps 26  $\rightarrow$  129  $\rightarrow$  246  $\mu\text{m}$  — the same thin-where-intact / thick-where-spent law as every top design.
- **Essentially uniform pitch**: zmid = +7.7 mm of the  $\pm 150$  allowed — the pitch knob contributed nothing here.
- Siblings R08\_00/R14\_01 independently confirm 4.37. A ceiling map, not a deployment candidate.

# Anatomy of the damage-free point

foilsfpfbp04R00\_02 — 49 foils, extent 1067 mm ( $\pm 133$  mm vs deployed), thickness  $\times 30$



| design            | $S/\sqrt{B}$ | damage      |
|-------------------|--------------|-------------|
| bp04R00_02        | <b>3.87</b>  | <b>0.59</b> |
| deployed default  | 3.26         | 0.69        |
| <i>difference</i> | +19%         | -13%        |

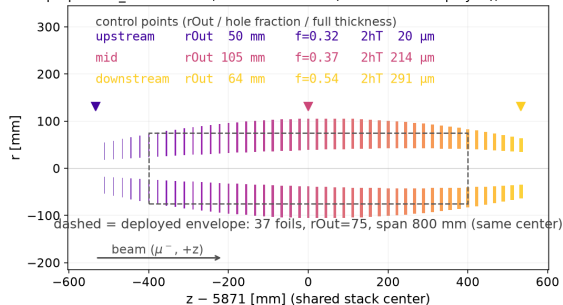
damage in  $10^{-6}$  MeV/POT, Run1Bap stack

- **Better on both objectives** than the target we deploy today — no trade at all. **24 of 83** evaluated designs do this.
- **A wider bore, and one that varies.** The deployed target is already annular, but at a *constant* 21.5 mm hole; this one runs **33–69 mm** — 1.5–3 $\times$  wider — and profiles it along  $z$ .
- Carries **216 cm<sup>3</sup>** of aluminium — 3.4 $\times$  the deployed 63 cm<sup>3</sup> — but only **1.5%** of it inside  $r < 40$  mm, against the deployed target's 22%.
- More stopping mass *and* less flash: the mass moved off the beam axis.



# Anatomy of the best point at deployed-level damage

foilsfpbp02R08\_00 — 49 foils, extent 1067 mm ( $\pm 133$  mm vs deployed), thickness  $\times 30$



| design            | $S/\sqrt{B}$ | damage     |
|-------------------|--------------|------------|
| bp02R08_00        | <b>4.05</b>  | 0.73       |
| deployed default  | 3.26         | 0.69       |
| <i>difference</i> | <b>+24%</b>  | <b>+6%</b> |

damage in  $10^{-6}$  MeV/POT, Run1Bap stack

- **The most significance available at deployed-level damage:** highest  $S/\sqrt{B}$  of any design within 10% of the default's flash.
- Buys +5 points of  $S/\sqrt{B}$  over the damage-free design for 19 points of damage — still only +6% against today's target.
- **Tighter bore:** 16–42 mm, straddling the deployed 21.5 mm, vs the damage-free point's 33–69 mm. Core fraction **3.1%** vs 1.5%.
- 212 cm<sup>3</sup> of Al,  $3.4\times$  the deployed 63 cm<sup>3</sup>, thickest downstream (291  $\mu\text{m}$ ) where the beam is spent.

# What actually drives the damage — everything vs. the deployed target

| design                          | $S/\sqrt{B}$ | vs default | damage      | vs default | Al [cm <sup>3</sup> ] | Al in $r < 40$ mm |
|---------------------------------|--------------|------------|-------------|------------|-----------------------|-------------------|
| deployed default (37 foils)     | 3.26         | —          | 0.69        | —          | 63                    | 22.0%             |
| bp04R00_02 ( <b>dominates</b> ) | 3.87         | +19%       | <b>0.59</b> | −13%       | 216                   | <b>1.5%</b>       |
| bp02R08_00                      | 4.05         | +24%       | 0.73        | +6%        | 213                   | 3.1%              |
| bp02R03_00                      | 4.09         | +25%       | 0.79        | +15%       | 166                   | 9.2%              |
| bp03R03_00                      | <b>4.30</b>  | +32%       | 0.84        | +22%       | 206                   | 9.6%              |
| bp01R00_03                      | 4.29         | +32%       | 1.23        | +79%       | 153                   | 22.0%             |

damage in  $10^{-6}$  MeV/POT; all measured on the Run1Bap stack; rows ordered by damage. Table and correlations computed on the 83-eval fixed-pitch (bp) set; the later 174 evals (box, pitch, exploit) leave the placement-not-mass conclusion unchanged.

Every optimized design carries **2.4–3.9×** the default's aluminium — that is how it buys the significance. What varies by **13×** is the damage *penalty*: −13% to +79%. It tracks the core column, not the mass column.

Over all **83** evaluated geometries:

| Spearman vs. damage                        | $\rho$       |
|--------------------------------------------|--------------|
| total Al volume                            | +0.57        |
| <b>Al inside <math>r &lt; 40</math> mm</b> | <b>+0.84</b> |

- bp01R00\_03 tripled the mass and *kept* the default's 22% core fraction → +79% damage. bp04R00\_02 carried *more* mass at a 1.5% core → −13%.
- So the lever is **placement, not mass** — and it is strong enough that **23 of 81** evaluated designs beat the deployed target on *both* objectives at once.
- Consistent with Edmonds (DocDB-10898) — flash parents are central,  $RMS \approx 24$  mm — found here by search, not by design.

# Why the optimum moves with the pitch

The same shape is *not* optimal at every length — transplanting the 2000 mm champion into this envelope leaves performance on the table.

| shape at 1066–1100 mm                       | $S/\sqrt{B}$ | hole profile $f$                        |
|---------------------------------------------|--------------|-----------------------------------------|
| transplanted 2000 mm champion (SCANA1100)   | 4.16         | 0.43 / 0.12 / <b>0.86</b>               |
| optimized at this pitch (bp01R00_03)        | 4.29         | 0.43 / 0 / <b>0</b>                     |
| optimized at this pitch (bp03R03_00)        | 4.30         | <b>0</b> / <b>0.17</b> / <b>0.19</b>    |
| exploit record (bpz04R11_00, uniform pitch) | 4.37         | 0.39 / <b>0.14</b> / <b>0.10</b>        |
| <b>record</b> (bpz05R01_00)                 | <b>4.41</b>  | <b>0.01</b> / <b>0.05</b> / <b>0.01</b> |

- +6% ( $\gg 5\sigma$  even at a conservative  $\sigma_{\text{sob}} = 0.6\%$ ) purely from re-optimizing the shape at fixed length. **Four independent optima found here all shut (or nearly shut) the downstream hole** the 2000 mm champion opens wide.
- **Mechanism:** at tight spacing a conversion electron re-crosses more neighbouring foils on its way out, so exit-path self-absorption dominates and *open holes stop paying*. At 42 mm pitch the same optimizer opens them wide.
- Consistent with the extent scan, where the length gain itself came from the CE low-momentum tail retreating — a narrower signal window against a cosmic-dominated background.

# What we tried — four parameterizations, one stopping rule

| campaign | axis opened                                              | evals | best $S/\sqrt{B}$ | best <i>inside</i> budget |
|----------|----------------------------------------------------------|-------|-------------------|---------------------------|
| bp01--04 | 9D shape at the deployed pitch                           | 80    | 4.33              | <b>4.00</b> (bp04R00_09)  |
| bw01     | bore/width knobs, $r_{\text{out}} \leq 150$ (cold start) | 20    | 3.79              | 3.79                      |
| bp01     | $r_{\text{out}}$ box 50–120 $\rightarrow$ 30–150         | 40    | 4.31              | 3.74                      |
| bpz01,03 | + $z_{\text{mid}}$ : non-uniform foil pitch              | 80    | 4.25              | 3.90                      |
| bpz04    | $\max S/\sqrt{B}$ <i>exploit</i> (pareto_sob picker)     | 40    | 4.37              | — (0/40 in budget)        |
| bpz05    | <i>exploit</i> , round 2                                 | 40    | <b>4.41</b>       | — (0/40 in budget)        |

budget = deployed damage,  $6.85 \times 10^{-7}$  MeV/POT. bpx/bpz warm-start on all prior rows, so their column shows what the *new* axis added — none displaced the incumbent 4.00. 317

evaluations total, incl. 13 transplanted seeds.

- **Stopping rule, fixed before each launch:** clear  $S/\sqrt{B} = 4.00$  *inside* the deployed damage budget, or the axis closes.
- **Shape** stalled at exactly 4.00 (bp04). **Box:** the front pressed the 120 mm  $r_{\text{out}}$  wall, but opening it to 150 (bpx, one variable changed) did not move best-at-budget — the wall was not binding. (bw had moved ceiling *and* parameterization together: negative but unattributable, hence the one-variable retry.) **Spacing** (bpz, first non-uniform-pitch search) did not clear it either. **Deployment verdict: 4.00, search closed 2026-08-08.**
- **The ceiling is budget-shaped, not box-shaped:** just past the line the front continues smoothly — **4.10** at +1% damage, 4.30 at +22%, **4.41** at +60%. More significance is on sale whenever more damage is acceptable.
- **The scalar "ceiling" was picker-relative:** after hypervolume acquisition saturated at 4.33, two exploit rounds (submit the GP's top-predicted-sob points) averaged **4.29 then 4.32 over their picks** and drove the record 4.33  $\rightarrow$  4.37  $\rightarrow$  **4.41** ( $\sim 5\sigma$  steps) — still climbing when stopped. Critical path is full-sim validation, not more BO.

# Conclusion & next steps

- A target optimized **inside the deployed envelope** gives +23%  $S/\sqrt{B}$  at the deployed damage budget (4.00 vs 3.26), and at least +35% (**4.41**) if +60% damage is acceptable — an envelope the existing support structure was built for.
- **A no-cost option exists.** bp04R00\_02: +19%  $S/\sqrt{B}$  at **-13%** damage — strictly better than the deployed target on both objectives, at the deployed foil pitch. This is the point to take to full simulation first. (42 of 297 designs beat the deployed target on both axes.)
- **Needed — prove it with full simulation.** Every number here is a fast score ( $S/\sqrt{B}$  and damage extracted without full reconstruction). The best configs must go through the full dig  $\rightarrow$  mcs  $\rightarrow$  nts chain before this is an  $R_{\mu e}$  claim.
- **Regime caveat.** Run-1a is cosmic-dominated, so part of the gain is window-economics. Shape gains (stopping  $\times$  acceptance) survive any background model; length gains do not.
- **Design lever, transferable:** damage tracks *core* aluminium ( $\rho = +0.84$ ), not total (+0.57) — keep the mass, move it off the axis.
- **Search closed for deployment** (317 evals: shape, widened box, pitch profile all stall at 4.00 in budget). The unconstrained ceiling is  $\geq$  **4.41 and still rising** (+0.04 per exploit round); its top is degenerate — a near-solid barrel at ramped pitch (4.41) and a funnel bore at uniform pitch (4.37) sit within 1% of each other.